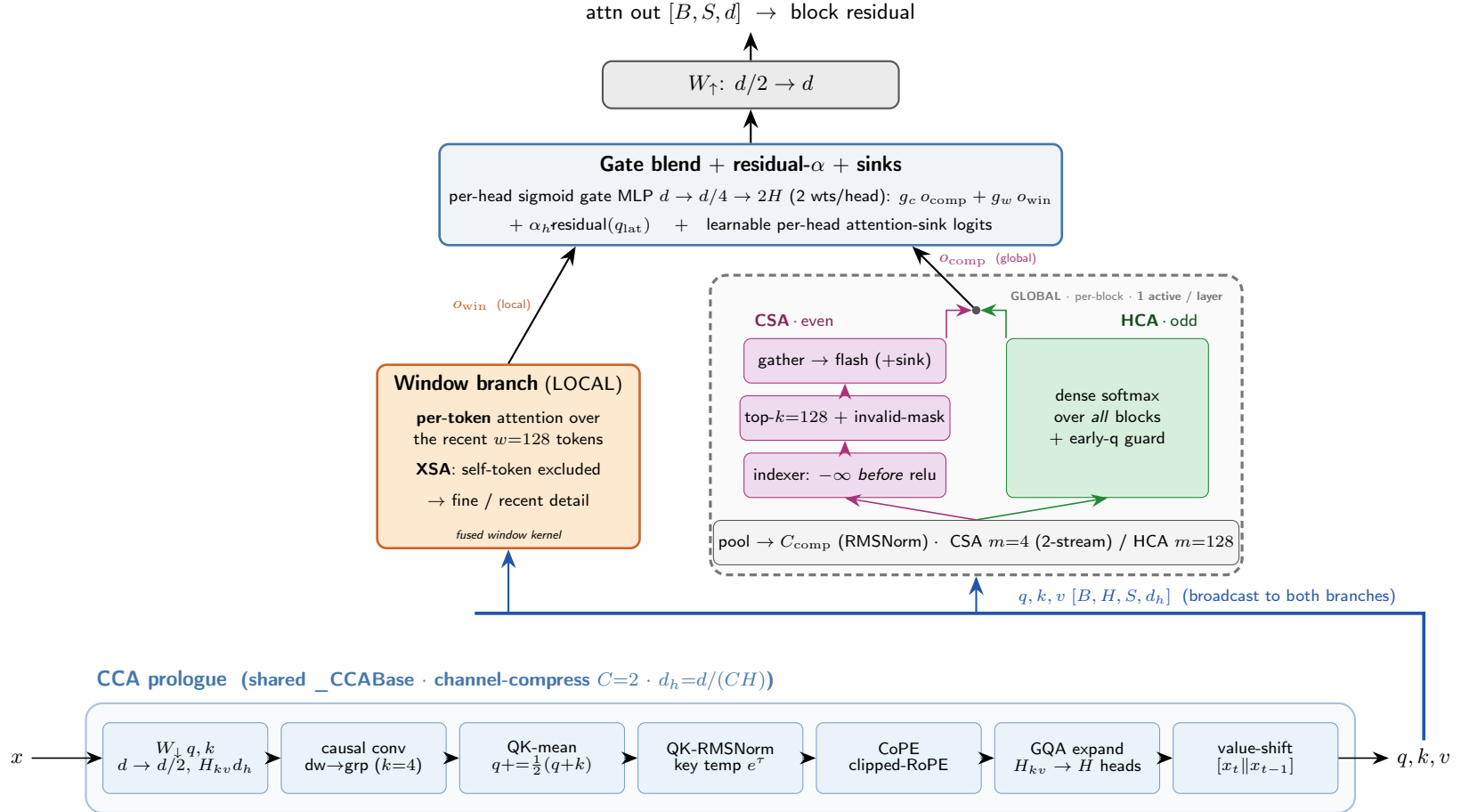


# MORPH Attention — Triple-Axis Compression

a shared CCA channel-compress front-end feeds a local window || one global-compressed branch (CSA / HCA), blended by a per-head gate



**Four compression axes** (faithful to morph/model/attention.py): channel CCA  $E \rightarrow E/2$  select CSA top-128 of  $S/4$  blocks dense HCA all  $S/128$  blocks local window  $w=128$ .  
**CSA/HCA alternate** on the *global* layer index (even $\rightarrow$ CSA, odd $\rightarrow$ HCA), baked at `__init__` — a looped core layer keeps its variant on every iteration (*not* iteration-conditioned).  
**Fused Triton**: CCA prologue, conv, window, CSA, HCA (each fwd/grad-exact). **Notation**:  $d$  model dim,  $H$  q-heads,  $H_{kv}$  kv-heads (GQA), channel-compress  $C=2$ ,  $d_h=d/(CH)$ , latent  $q=d/2$ ,  $n_{blk}=S/m$ . Color secondary — every box is text-labeled.